

MOHAMMED SHAMIL K P

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SUMMARY

Electronics and Communication Engineering student with hands-on experience in robotics system design, embedded programming, and hardware–software integration. Skilled in solving technical challenges and passionate about applying engineering knowledge to real-world problems. Active participant in hackathons, innovation projects, and community initiatives.

TECHNICAL SKILLS

Programming & Tools: Python, MATLAB, Embedded C, Arduino, Robotics Hardware, Microcontrollers

Domains: Robotics, Embedded Systems, VLSI Design, Data Science, AI/ML Fundamentals

Languages: English, Malayalam

EXPERIENCE

Motridox Robotics

Embedded Systems and Hardware Lead — Student Robotics Group

Sep 2022 – Present

Malappuram, Kerala

Designed and assembled electronic hardware for custom robotic prototypes.

Developed and tested embedded programs for microcontrollers and sensors.

Integrated hardware and software to achieve functional robotic systems.

Prepared concise technical reports, presentations, and documentation.

Led and participated in hackathons, including NASA Space Apps Challenge.

Achieved 2nd place at Zilkathon Hackathon.

PROJECTS

FaceLytix – Intelligent Touchless Attendance System

Eranad Knowledge City Technical Campus, Cherukulam, Manjeri

Dec 2024 – Apr 2025

github.com/shamil1355/FaceLytix

Developed an intelligent touchless attendance system using **ESP32-CAM** and **OpenCV** for real-time facial recognition and logging.

Handled **hardware installation**, **software integration**, and complete **framework design**.

Implemented secure and automated attendance management suitable for schools and offices.

Technologies: ESP32, Raspberry Pi, OpenCV, Python, Arduino IDE, Documentation Tools.

Westo – Smart Waste Bin with Auto-Compression

IoT & Embedded Systems Project

Dec 2025 – Mar 2026

github.com/MotridoxRobotics/WasteBin

Developed a smart waste management system integrating **embedded systems**, **robotics**, and **mobile app control**.

Implemented automatic lid operation using a **servo motor** with proximity detection via ultrasonic sensors.

Designed a **dual stepper motor compression mechanism** using A4988 drivers for efficient waste handling.

Built real-time monitoring with Wi-Fi and a **web dashboard** with REST API integration.

Created a **Flutter mobile app** for monitoring and manual control of the compression system.

Technologies: ESP32, HC-SR04, A4988, Stepper Motors, Servo, Arduino, Flutter, REST API.

The-V Autonomous Waste Management Robot

Robotics & AI Integration Project

Ongoing

Designing and developing an **autonomous waste collection and sorting robot** to improve efficiency in urban waste management systems.

Implemented **camera-based navigation** using computer vision for real-time path planning and obstacle avoidance in dynamic environments.

Integrated a **robotic arm** for automated waste pickup along with an intelligent sorting mechanism to classify waste based on material type.

Developed an **IoT-enabled monitoring system** providing real-time data on robot status, waste levels, and location through a cloud-based dashboard.

Designed an **automatic docking system** for self-charging and waste disposal, reducing human intervention.

Focused on building a **sustainable, scalable, and intelligent solution** combining robotics, AI, and IoT technologies.
Technologies: Computer Vision, AI/ML, IoT, Robotics, Embedded Systems, Cloud Integration.

EDUCATION

Eranad Knowledge City Technical Campus <i>B.Tech in Electronics and Communication Engineering</i> Comprehensive program focusing on core electronics, communication theory, and signal processing. Active member of NSS, IEDC , and IEEE Student Branch . Skilled in 8051 Microcontroller, HDL, Embedded C, Python, and system design. Participated in seminars, workshops, and research-based innovation projects.	Manjeri, Kerala <i>Sep 2022 – Oct 2026</i>
Darul Uloom Higher Secondary School (HSS) Panakkad <i>Higher Secondary, Computer Science</i> Specialized in Computer Science and Information Systems with focus on programming fundamentals. Coursework included programming languages, DBMS, and web development with practical lab sessions. Gained hands-on experience through coding projects and interschool technical events. Built a solid foundation for engineering and problem-solving skills.	Kerala, India <i>Jun 2020 – Mar 2022</i>

PUBLICATIONS

The V: Autonomous Waste Collection Robot <i>International Journal of Creative Research Thoughts (IJCRT)</i> Paper ID : IJCRT2603495 Published a research paper on an autonomous waste collection and segregation robot integrating computer vision, robotics , and IoT-enabled monitoring . Implemented OpenCV-based waste classification, robotic arm control, autonomous navigation, and intelligent waste segregation. Technologies: OpenCV, Raspberry Pi 4, Robotics, IoT, Computer Vision, Embedded Systems.	Mar 2026 <i>Published</i>
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CERTIFICATIONS

Artificial Intelligence – NoviTech R&D Pvt Ltd (2025)
Project to Product – Eranad Knowledge City (2024)
VLSI System on Chip Design – Maven Silicon (2024)
Enterprise Data Science – IBM (2024)
Scientific Computing & Data Science Using Python – EKC (2023)

VOLUNTEERING

National Service Scheme (NSS) – Contributed to community welfare, cleanliness drives, and awareness programs.
Indian Red Cross Society (JRC Cadet) – Engaged in youth empowerment and health-related initiatives (2018–2020).